



ELE466 - Machine Learning



HANDWRITTEN LETTERS RECOGNITION USING CNN

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WHAT IS THE PROJECT ABOUT?

This project builds a machine learning system to recognize handwritten English letters (A–Z) using a Convolutional Neural Network (CNN).

- Input: 28×28 grayscale images of letters
- Goal: Classify each image as one of the 26 English letters
- Dataset: A–Z Handwritten Letters from Kaggle (over 300,000 samples)
- Achieved high accuracy: 99.46% on test data
- Can be used in applications like: documents digitalization and form reading





METHODOLOGY - THEORETICAL BACKGROUND

The model uses a Convolutional Neural Network, which is ideal for image classification. CNNs learn patterns by detecting spatial features in images.

- Conv2D: Extracts local features (like edges, curves)
- MaxPooling2D: Reduces feature map size, keeps essential info
- Flatten & Dense: Combines features to classify the letter
- Dropout: Prevents overfitting by randomly disabling neurons
- Softmax Output: Predicts the correct letter from A-Z (26 classes)





METHODOLOGY - IMPLEMENTATION

- **Preprocessing:**

- Images normalized (values scaled to 0–1)
- Reshaped to 28×28×1 (grayscale format)
- Labels converted to one-hot encoding

- **Training Details:**

- 20 epochs, batch size 128
- Used Google Colab for free GPU acceleration
- Model saved for later prediction





RESULTS



Test Accuracy: 99.46%

- 15 out of 26 letters had F1-scores above 99.5%
- Best performer: "S" (99.9%)
- Most confusion: "D" misclassified as "O" (3.7% error)
- Speed: 9 milliseconds per image (~111 FPS)
- Model size: 2.3 MB (lightweight)

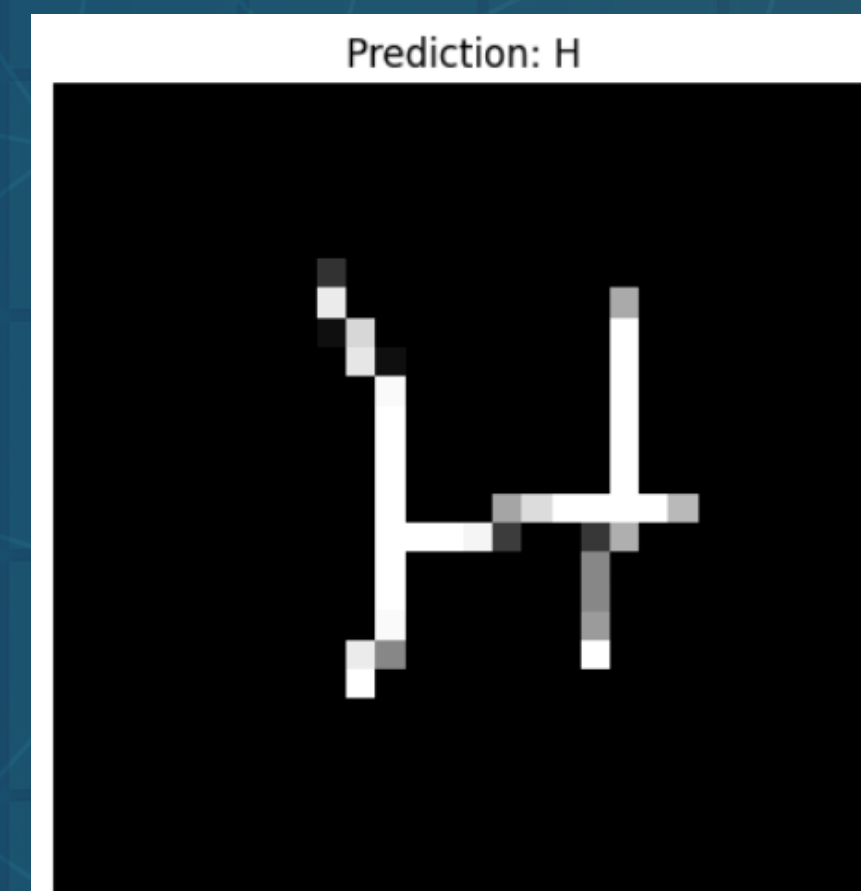




SAMPLE PREDICTIONS



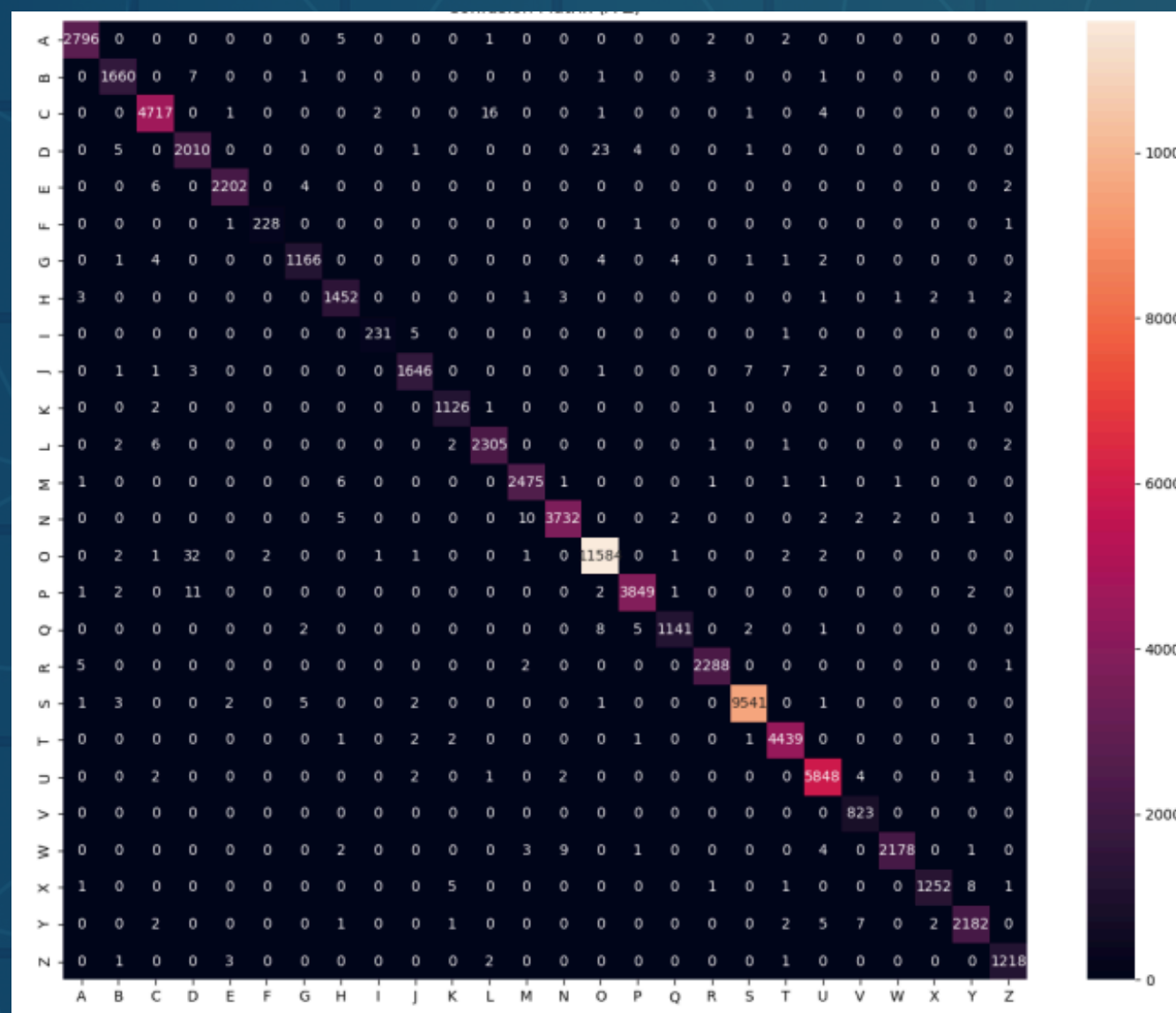
Pred: J True: J	Pred: O True: O	Pred: C True: C	Pred: A True: A
			
Pred: U True: U	Pred: A True: A	Pred: Y True: Y	Pred: D True: D
			
Pred: O True: O	Pred: S True: S	Pred: S True: S	Pred: G True: G
			





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CONFUSION MATRIX HEATMAP





CONCLUSION

The CNN-based system accurately recognized isolated handwritten letters with near-perfect accuracy and fast performance.

1.

Strengths

- High precision across most letter types
- Small model size, fast prediction
- Easy to implement and deploy

2.

Limitations

- Struggles with similar curved shapes (e.g., D vs. O)
- Only works for isolated characters not cursive or words





THANK YOU!

Any Questions?

